

Bending

Shear / Moment Diagram

$$\textcircled{1} \quad A_x + T = 0$$

$$\textcircled{2} \quad A_y - V = 0$$

$$\textcircled{3} \quad M - Vx = A_y \cdot x$$

$$A_y - P - V = 0$$

$$M - Vx - P \cdot b = 0$$

$$\Rightarrow V = A_y - P$$

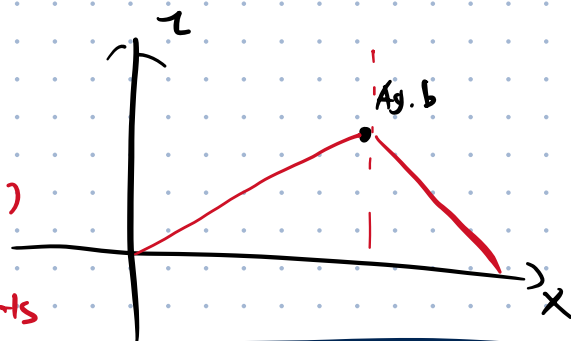
$$M = Px - A_y x - Pb$$

$$\textcircled{0} \quad V_0 = F_0 \text{ (该点外力)}$$

$$\textcircled{1} \quad \frac{dV}{dx} = w(x) \text{ (如果是连续力)}$$

$$\textcircled{2} \quad \frac{dM}{dx} = V(x) \quad (V \rightarrow 0, M \text{ reaches max})$$

$\textcircled{3} \quad V$ is discontinuous at points where a concentrated load.

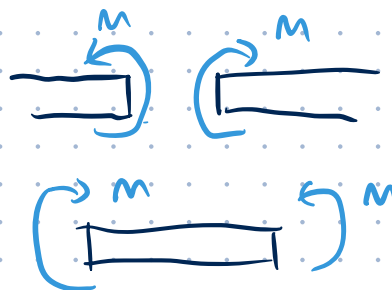
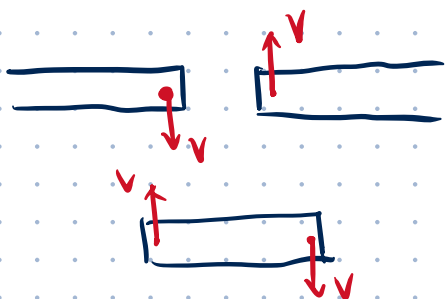


Jump = 这个点的外力 (大小, 正负号同)

$\textcircled{4} \quad M$ is discontinuous at points where there's a concentrated moment.
The discontinuity in M is negative of the concentrated moment

$$\text{Jump} = - \text{该点 Moment}$$

Sign Conversion



Bending Deformation of a Straight Member

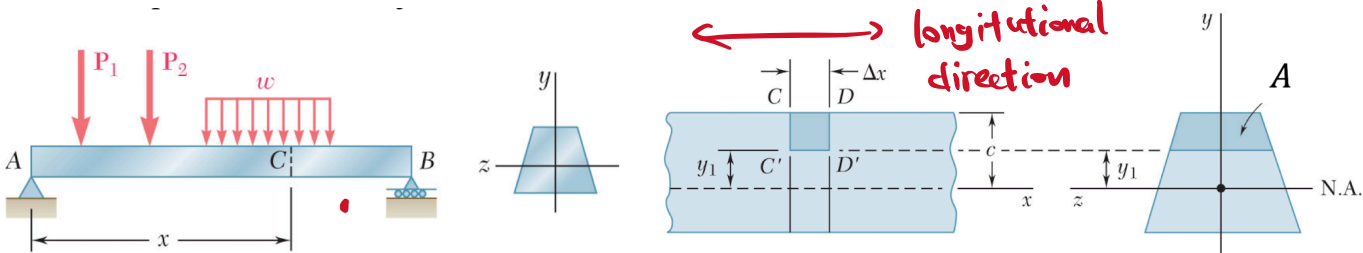
$$\sigma_x = \frac{-My}{I}$$

$$\sigma_{max} = \frac{Mc}{I}$$

★ compressive

还是 Tensile !!!

Shear Flow



$$\sigma = -\frac{My}{I_z}$$

$$\sum F_y = 0 \Rightarrow V - V = 0$$

$$\sum F_x = 0 \Rightarrow$$

$$-\Delta H + \int \sigma_c dA + \int \sigma_b dA = 0$$

$$-\Delta H + \int_{c'} -\frac{M_c}{I_z} y dA - \int_{b'} -\frac{M_b}{I_z} y dA = 0 \leftarrow$$

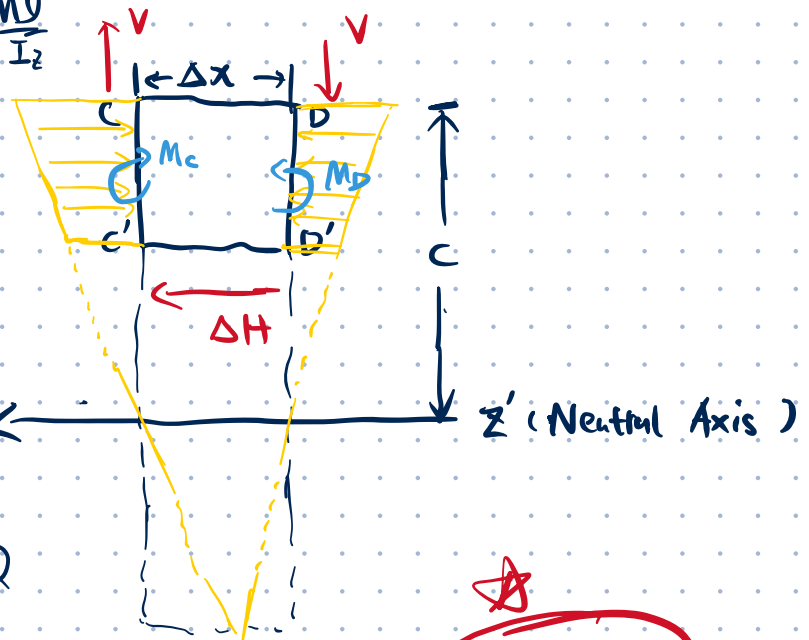
$$\Delta H = \frac{M_b - M_c}{I_z} \int y dA = \frac{(M_b - M_c)}{I_z} Q$$

$$Q = \int y dA = \sum \bar{y} A$$

where $\bar{y}$: 小块块的 centroid
A: 小块块的 面积

★ to NA

不能直接用 y 值!



$$\Rightarrow \frac{\Delta H}{\Delta x} = \frac{M_D - M_C}{\Delta x I_z} Q \Rightarrow$$

$$\text{Shear Flow } q = \frac{VQ}{I_z} \text{ [N/m]}$$

force per unit length (longitudinal)

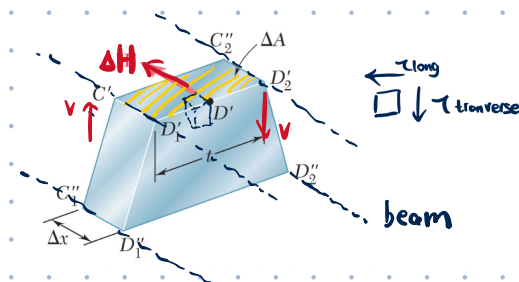
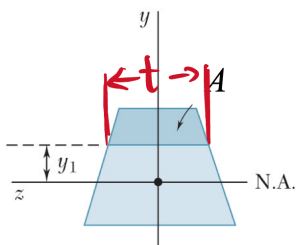
$$\left\{ \begin{array}{l} \frac{M_D - M_C}{\Delta x}, \frac{dM}{dx} = V \\ q = \frac{dH}{dx} \text{ shear flow} \end{array} \right.$$

$$\text{Average Shear stress } \tau_{\text{long}} = \frac{q}{t} = \frac{VQ}{I_z t}$$

where t : 连接处长度

$\Delta \tau_{\text{max}}$ at N.A.?

τ_{avg} along direction of t 连接处求均值



Built Up Beam With Fasteners



ΔH is Force longitudinal

$$dH = q dx$$

$$F_{\text{long}} = q \Delta x \rightarrow F_{\text{nail}}$$

$$F_{\text{nail}} = q \Delta S = \tau_{\text{nail}} A_{\text{nail}}$$

Where ΔS : Fastener 间距

τ_{nail} : 钉子的 shear stress (剪切)

A_{nail} : 钉子被截面积

* 注意 n 个钉子, $A_{\text{nail}} = \sum_n A$

此处求 q 技巧

① Q 要求被钉子分割的个块块